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Project Writeup

Problems Description:

The game snake is a single-user game where players use arrow keys to collect apples. Snake is played on a 15 x 15 grid; if the snake either exits the board runs into itself, the game ends. The snake starts off small, and as each apple is consumed, the length of the snake increases. The object of the game is to collect as many apples as possible, eventually filling the entire board with the snake. Our goal is to implement a RL algorithm which efficiently completes the game with high probability.

Dataset Source: Snake Game

There is no need for a dataset per say, but this assumes that we will be learning all game play state transitions from scratch. Yet this may not be the best solution, we can gather our own dataset by manually playing the game and initializing the Q-learning agent with prior knowledge instead of starting from scratch with random values or zeros. This idea is known as warm RL and circumvents the tedious process of having to learn innate movements that us humans deem to be trivial. We will be comparing how random initialization and this "pre-training" step compares to each other in game output quality as the iterations of the training increase.

Methodology: Approximate Q-Learning

Even though there is a perfect algorithm to solve Snake such as the ‘Hamiltonian Cycle’, we will add extra features to prevent the snake from taking a long way around.

We are going to use

- Distance to wall, apple
- Current length of the snake
- Current number of steps that the snaked moved

To evaluate the score.